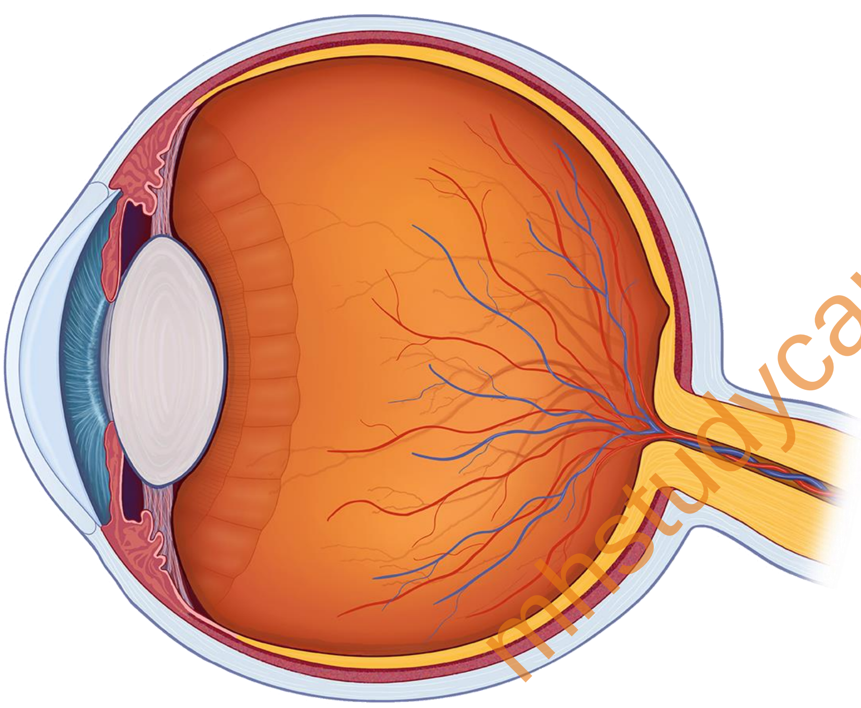
Chapter 7

LENSES



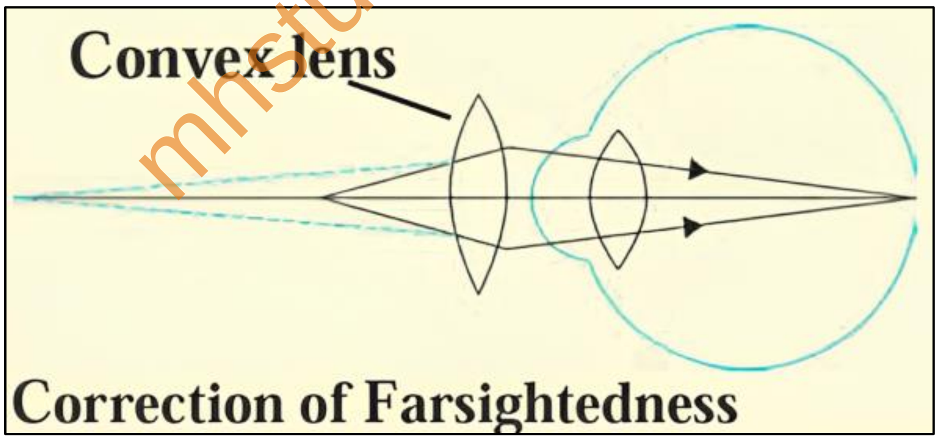
Question And Answers

1. Match the columns in the following table and EXPLAIN them.

Column 1	Column 2	Column 2
Farsightedness	Far away object can be seen clearly	Convex lens
Presbyopia	Problem of old age	Bifocal lens
Near-sightedness	Nearby object can be seen clearly	Concave lens

EXPLANATION:

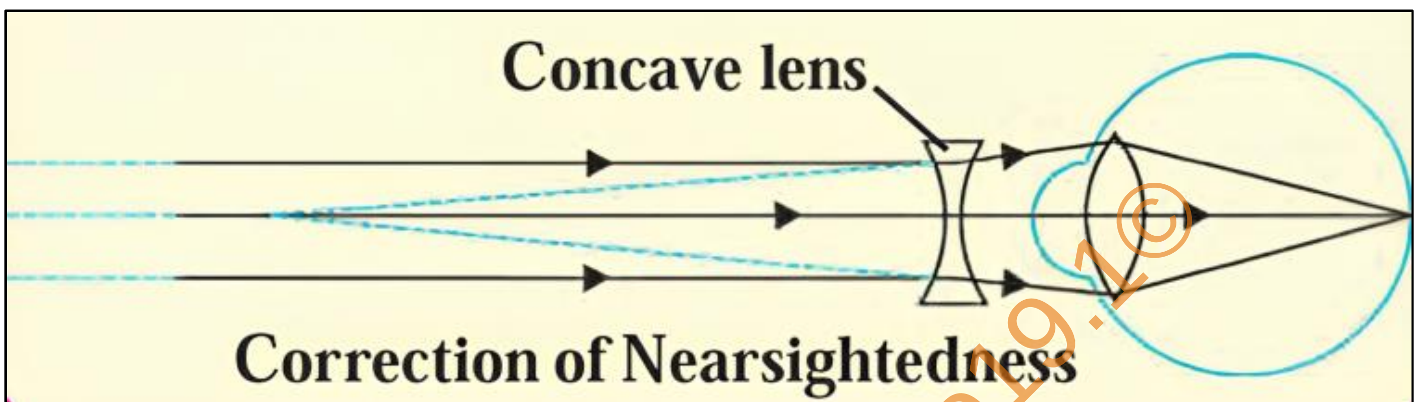
i. Farsightedness is caused due to reduction in the curvature of the cornea and eye lens. The converging power of the lens becomes less. In this defect the human eye can see distant objects clearly but cannot see nearby objects distinctly. Since a convex lens can converge incoming rays, it can be used to correct this defect of eye.



Just for reference.

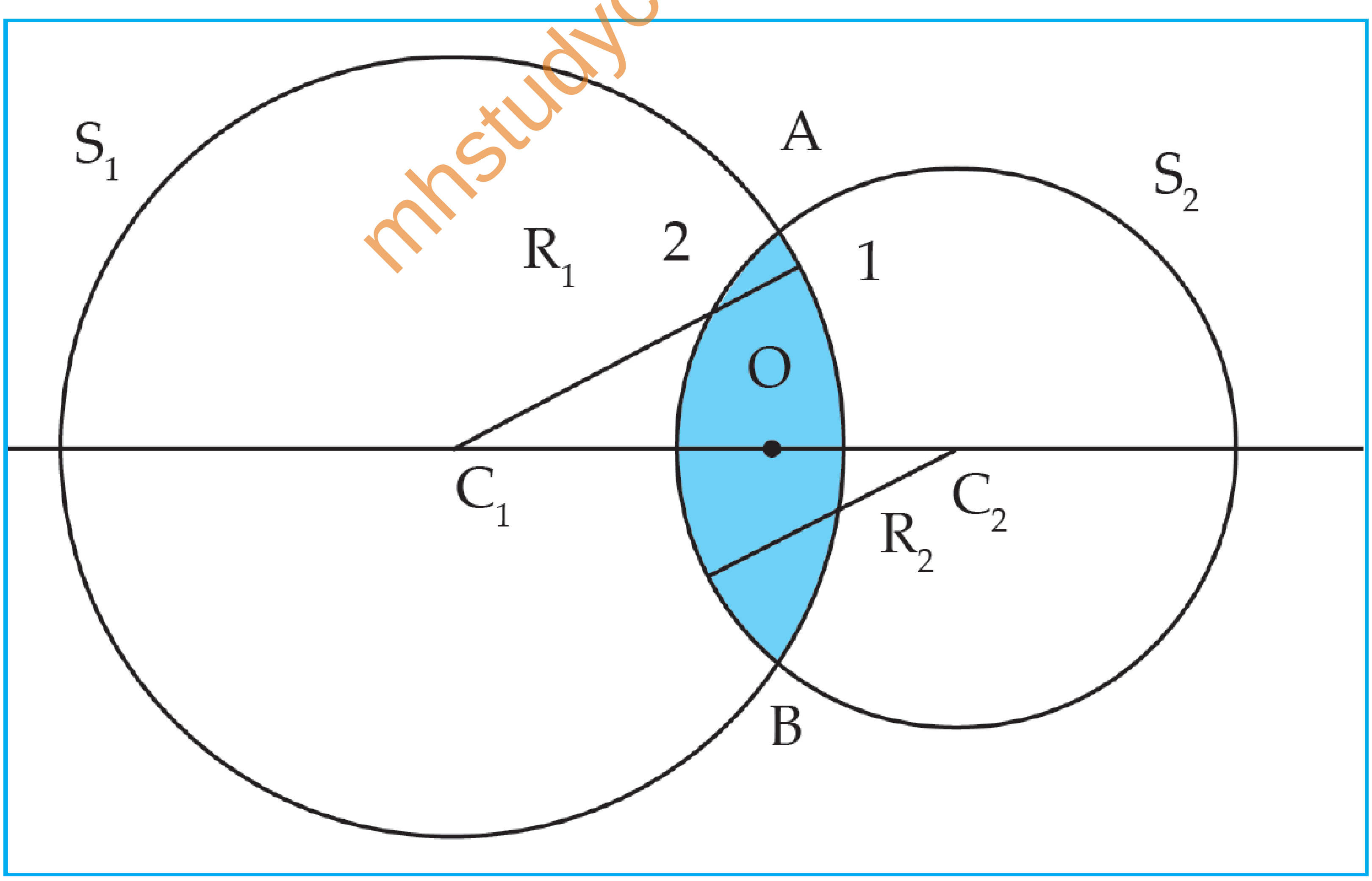
ii. The focusing power of the eye lens decreases with age. The muscles near the lens lose their ability to change the focal length of the lens. Old people suffer from nearsightedness as well as farsightedness. This condition is called as Presbyopia. In such a case bifocal lenses are required to correct the defect.

iii. Nearsightedness is caused due to increase in the curvature of the cornea and eye lens. The muscles near the lens cannot relax so that the converging power of the lens remains large. In this defect the human eye can see nearby objects clearly but cannot see distant objects distinctly. Since a concave lens can diverge incoming rays, it can be used to correct this defect of eye.

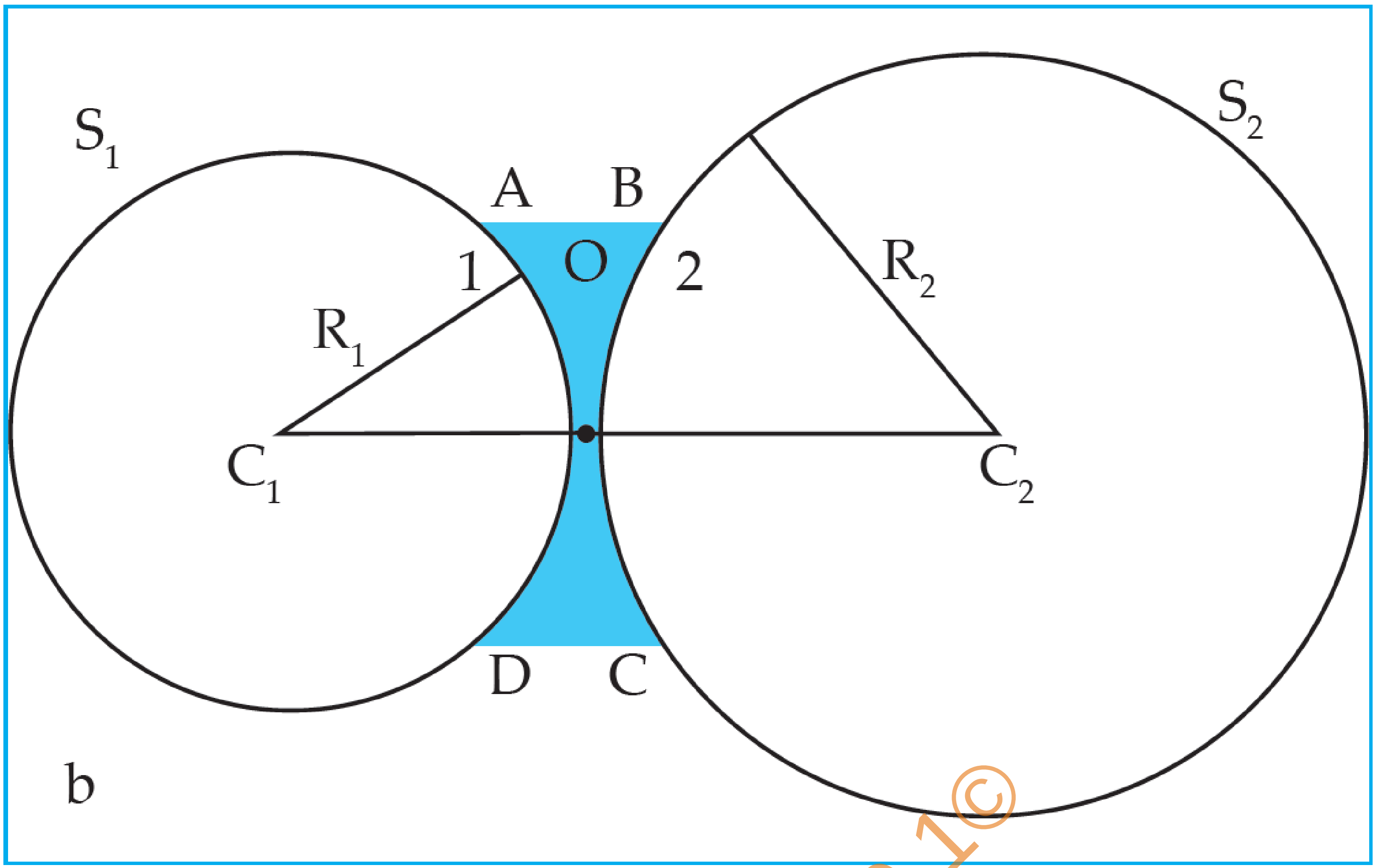


Just for reference.

2. Draw a figure explaining various terms related to a lens.



Cross section of convex lens



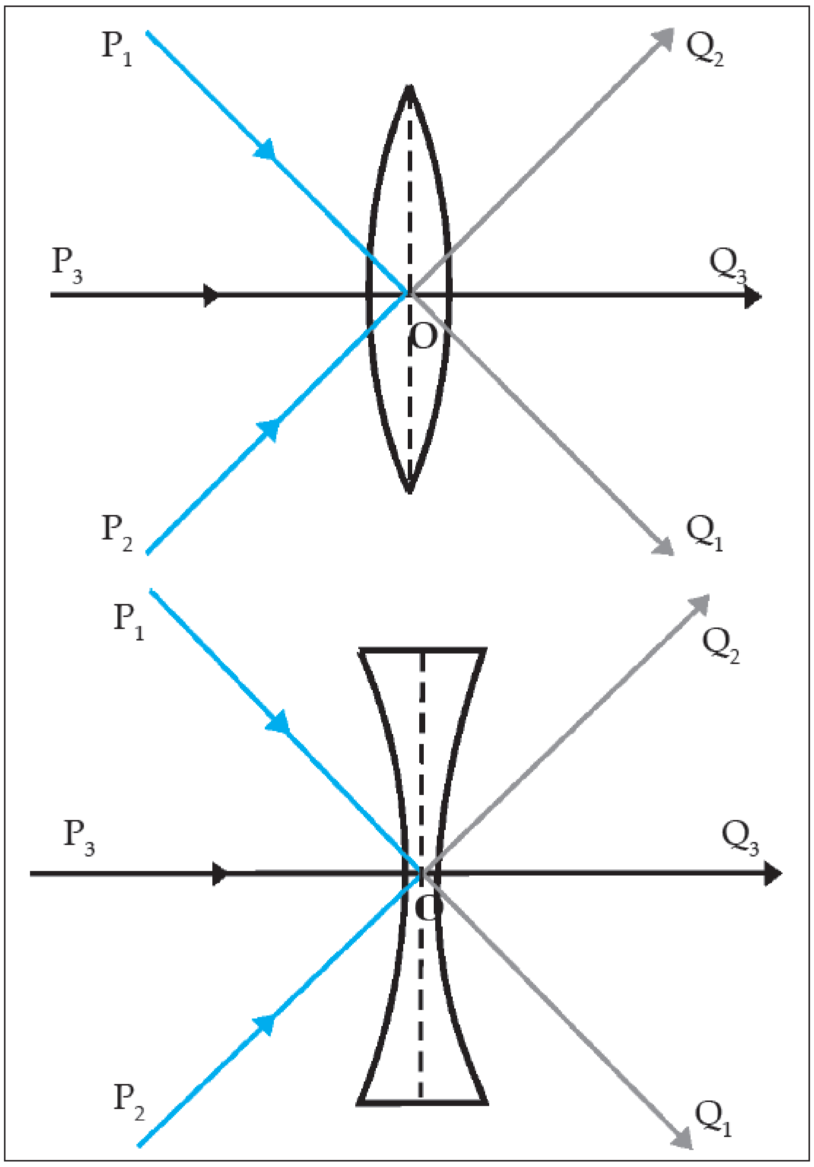
Cross section of concave lens

i. Centre of curvature (C): The centres of spheres whose parts form surfaces of the lenses are called centres of curvatures of the lenses. A lens with both surfaces spherical, has two centres of curvature C_1 and C_2.

ii. Radius of curvature (R): The radii (R_1 and R_2) of the spheres whose parts form surfaces of the lenses are called the radii of curvature of the lens.

iii. Principal axis: The imaginary line passing through both centres of curvature is called the principal axis of the lens.

iv. Optical centre (O): The point inside a lens on the principal axis, through which light rays pass without changing their path is called the optical centre of a lens. In the following figure, rays P_1Q_1, P_2Q_2 passing through O are going along a straight line. Thus, O is the optical centre of the lens.



Optical centre of a lens

(a) Principal focus of a convex lens

(b) Principal focus of a concave lens

v. Principal focus (F): When light rays parallel to the principal axis are incident on a convex lens, they converge to a point on the principal axis. This point is called the principal focus of the lens. As shown in above figure-a, F_1 and F_2 are the principal foci of the convex lens.

vi. Light rays parallel to the principal axis falling on a convex lens come together, i.e. get focused at a point on the principal axis. So, this type of lens is called a converging lens.

Rays travelling parallel to the principal axis of a concave lens diverge after refraction in such a way that they appear to be coming out of a point on the principal axis. This point is called the principal focus of the concave

i. lens. As shown in above figure-b, F_1 and F_2 are the principal foci of the concave lens.

ii. Light rays parallel to the principal axis falling on a concave lens go away from one another (diverge) after refraction. So, this type of lens is called a diverging lens.

vi. Focal length (f): The distance between the optical centre and principal focus of a lens is called its focal length.

3. At which position will you keep an object in front of a convex lens so as to get a real image of the same size as the object? Draw a figure.

When an object is placed at the centre of curvature $2F_1$ of a convex lens, we will get a real image of the same size as the object.

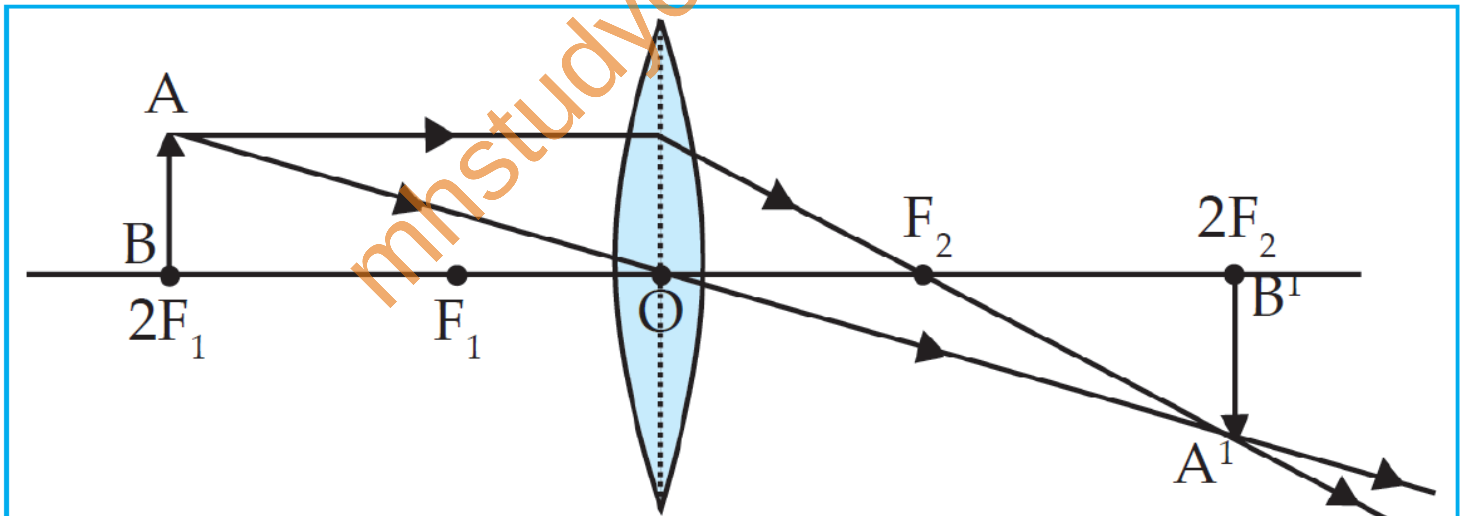


Image position: At $2F_2$

Nature: Real, inverted and of same size.

4. Give scientific reasons:

a. Simple microscope is used for watch repairs.

i. A convex lens with small focal length produces a virtual, erect and bigger image of an object.

ii. Such a lens is called simple microscope or magnifying lens.

iii. One can get a 20 times larger image of an object using such microscopes.

iv. Therefore, it is used by watch repairers to see the minute parts of the watch clearly without causing any strain to the eyes.

b. One can sense colours only in bright light.

i. The retina in our eyes is made up of many light sensitive cells. These cells are shaped like a rod and like a cone.

ii. The rod like cells respond to the intensity of light and give information about the brightness or dimness of the object to the brain. The conical cells respond to the colour and give information about the colour of the object to the brain.

iii. Brain processes all the information received and we see the actual image of the object.

iv. Rod like cells respond to faint light also but conical cells do not. Thus, we perceive colours only in bright light.

c. **We can not clearly see an object kept at a distance less than 25 cm from the eye.**

i. The muscles attached to the eye lens (ciliary muscles) help in fine adjustments of the focal length of the lens.

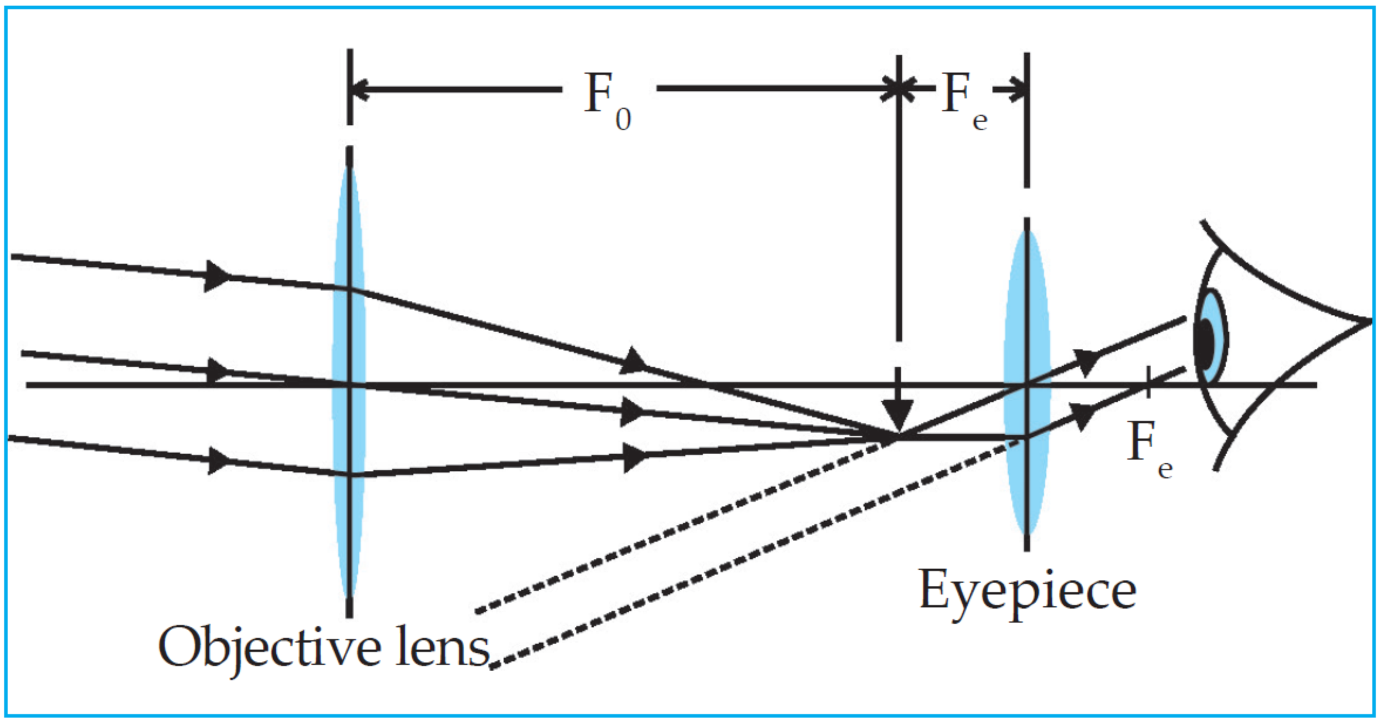
ii. The capacity of these muscles to contract or relax to adjust the focal length (i.e., power of accommodation) has a limit.

iii. The minimum distance of an object from a normal eye, at which it is clearly visible without stress on the eye, is called as minimum distance of distinct vision.

iv. The position of the object at this distance is called the near point of the eye, for a normal human eye, the near point is at 25 cm.

v. Hence, We can not clearly see an object kept at a distance less than 25 cm from the eye

5. Explain the working of an astronomical telescope using refraction of light.



To be drawn in exam.

i. Telescope is used to see distant objects clearly in their magnified form. The telescopes used to observe astronomical sources like the stars and the planets are called astronomical telescopes. Telescopes are of two types.

1. Refracting telescope – This uses lenses
2. Reflecting telescope – This uses mirrors and also lenses.

ii. In both, the image formed by the objective acts as object for the eye piece which forms the final image. Objective lens has large diameter and larger focal length because of which maximum amount of light coming from the distant object can be collected.

iii. On the other hand, the size of the eyepiece is smaller, and its focal length is also less.

iv. Both the lenses are fitted inside a metallic tube in such a way that the distance between them can be changed.

v. The principal axes of both the lenses are along the same straight line. Generally, using the same objective but different eye pieces, different magnification can be obtained.

vi. When rays of light enter the objective, they refract and give a real inverted and diminished image.

vii. The eye–piece is so adjusted that the image becomes an object for the eyepiece and gives a virtual, enlarged and inverted image w.r.t to object.

6. Distinguish between:

Farsightedness/ Hypermetropia	Near-sightedness/ Myopia
The curvature of the cornea and eye lens decreases so that the converging power becomes less.	The curvature of the cornea and eye lens increases so that the converging power becomes more.
Due to the flattening of the eyeball the distance between the lens and retina decreases.	The eyeball elongates so that the distance between the lens and the retina increases.
A person suffering from this defect can see distant objects clearly but is unable to see nearby objects.	A person suffering from this defect can see nearby objects clearly but is unable to see distant objects.
It can be corrected by using spectacles having convex lenses of suitable focal length.	It can be corrected by using spectacles having concave lenses of suitable focal length.

Concave lens	Convex lens
It is also called as diverging lens.	It is also called as converging lens.
This lens is thinner in the centre than at its edges.	This lens is thicker at the centre than at its edges.
The focal length is negative.	The focal length is positive.
This lens is used to correct myopia.	This lens can be used to correct hypermetropia.

7. What is the function of iris and the muscles connected to the lens in human eye?

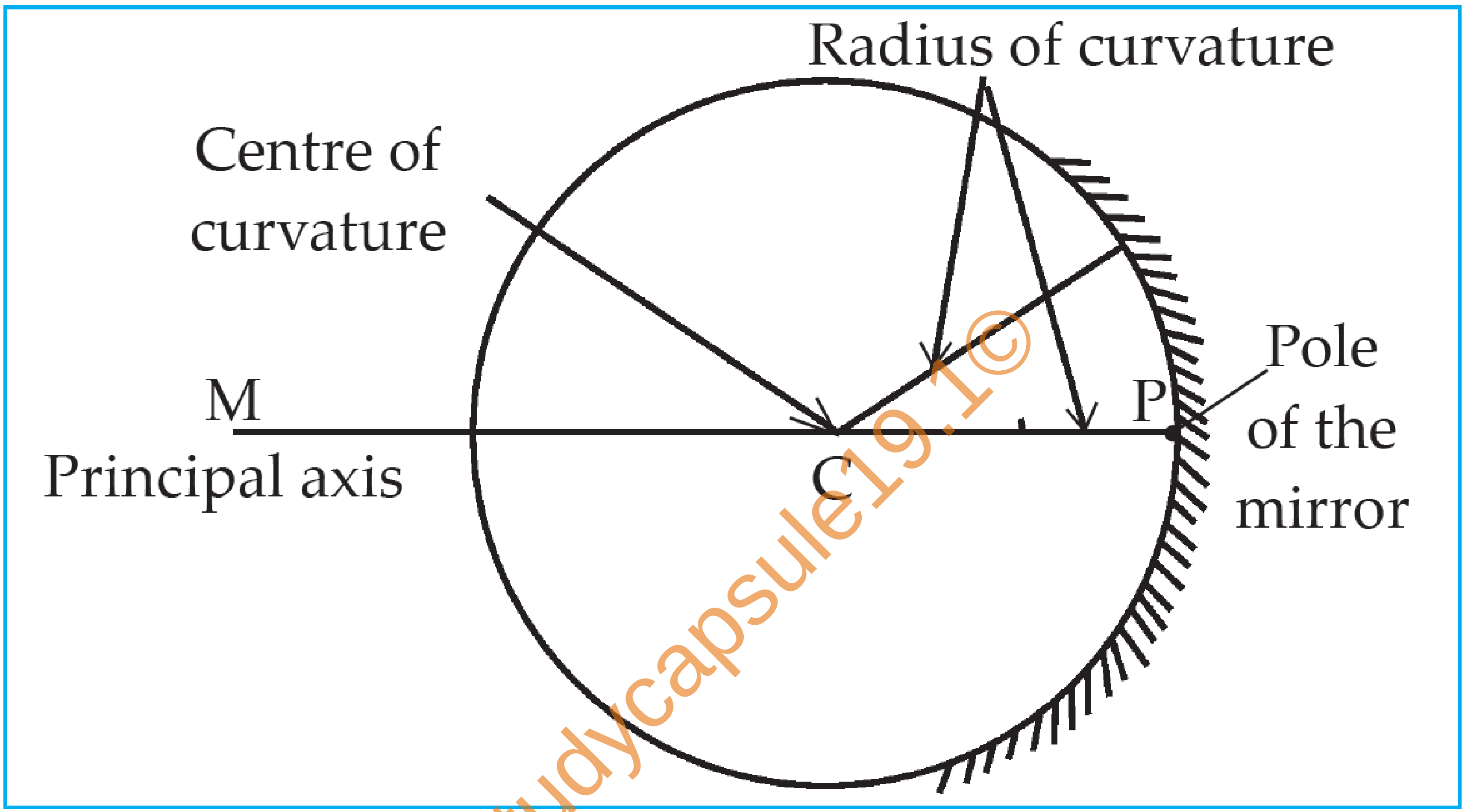
i. Iris gives the eye color. The key function of this is to regulate the amount of light that reaches the eye.

ii. Ciliary muscles (Linked to the eye lens) contract and relax to adjust the lens focal length. That helps us to see things nearby and far away.

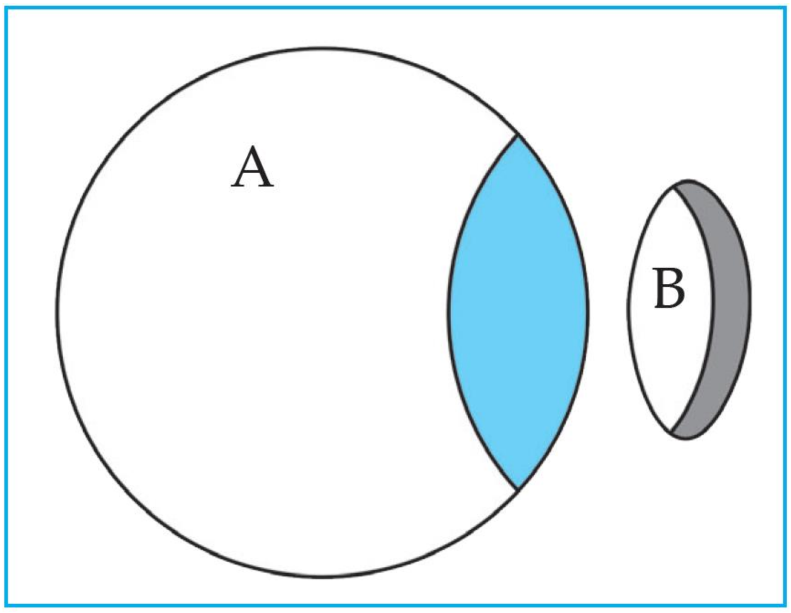
8. Solve the following examples. – At the back

INTEXT QUESTIONS

1. Indicate the following terms related to spherical mirrors in given figure: poles, centre of curvature, radius of curvature, principal focus.



2. How are concave and convex mirrors constructed?



i. Generally, concave and convex mirror are spherical mirrors.

ii. Spherical mirrors are parts of a hollow glass sphere like the part B in the figure.

iii. The inner or outer surface of this part is coated with a shiny substance to produce spherical mirror.

3. What are real and virtual images? How will you find out whether an image is real or virtual? Can a virtual image be obtained on a screen?

i. If the reflected or refracted rays actually meet at a point, then the image formed is Real and it can be seen on a screen.

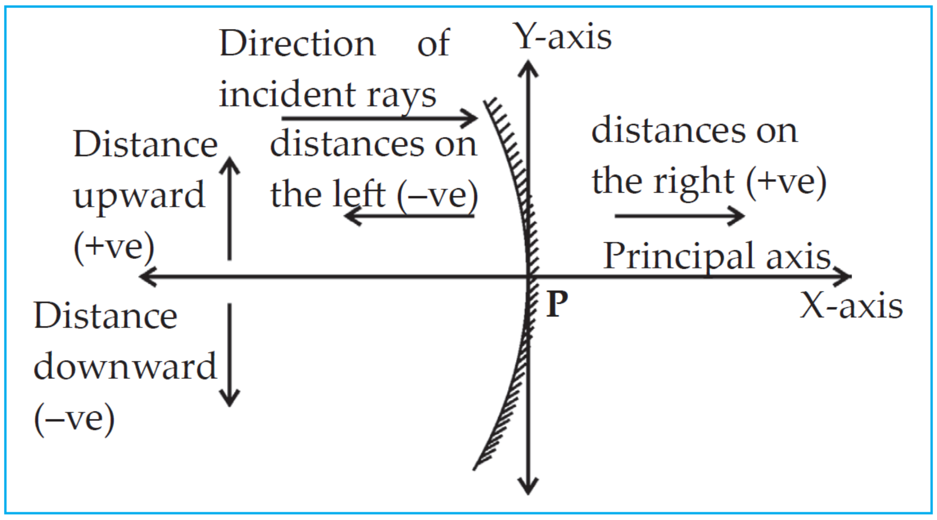
ii. If the reflected or refracted rays appear to meet at a point, then the image formed is called virtual.

ii. If the images are inverted, then they are real and if they are erect then they are virtual.

iii. Virtual images cannot be obtained on a screen.

4. What is the Cartesian sign convention used for spherical mirrors?

According to the Cartesian sign convention, the pole of the mirror is taken as the origin. The principal axis is taken as the X–axis of the frame of reference. The sign conventions are as follows.



i. The object is always kept on the left of the both mirrors. All distances parallel to the Principal axis are measured from the pole of the mirror.

ii. All distances measured towards the right of the pole are taken to be positive, while those measured towards left are taken to be negative.

iii. Distances measured vertically upwards from the principal axis are taken to be positive.

iv. Distances measured vertically downwards from the principal axis are taken to be negative.

v. The focal length of concave mirror is positive while that of a convex mirror is negative.

5. What is the relation between h_1, h_2, u and v?

The ratio of height of image (h_2) to the height of the object (h_1) is called Magnification (M).

$$\boxed{M = \frac{h_2}{h_1}} \quad ....(i)$$

Magnification can also be calculated by the ratio of the image distance (v) to the object distance (u).

$$\boxed{M = \frac{v}{u}} \quad ....(ii)$$

From (i) and (ii),

$$\boxed{M = \frac{h_2}{h_1} = \frac{v}{u}}$$

6. Why do we have to bring a small object near the eyes in order to see it clearly?

i. The Apparent size of an object depends on the angle subtended by the object at the eye.

ii. When the object is closes to the eye, the angle subtended is larger and the object appears bigger.

iii. Hence, we have to bring a small object near eye.

7. If we bring an object closer than 25 cm from the eyes, why can we not see it clearly even though it subtends a bigger angle at the eye? - Same answer as 4 – c

8. How do we perceive different colours?

i. The retina in our eyes is made up of many light sensitive cells. These cells are shaped like a rod and like a cone.

ii. The rod like cells respond to the intensity of light and give information about the brightness or dimness of the object to the brain.

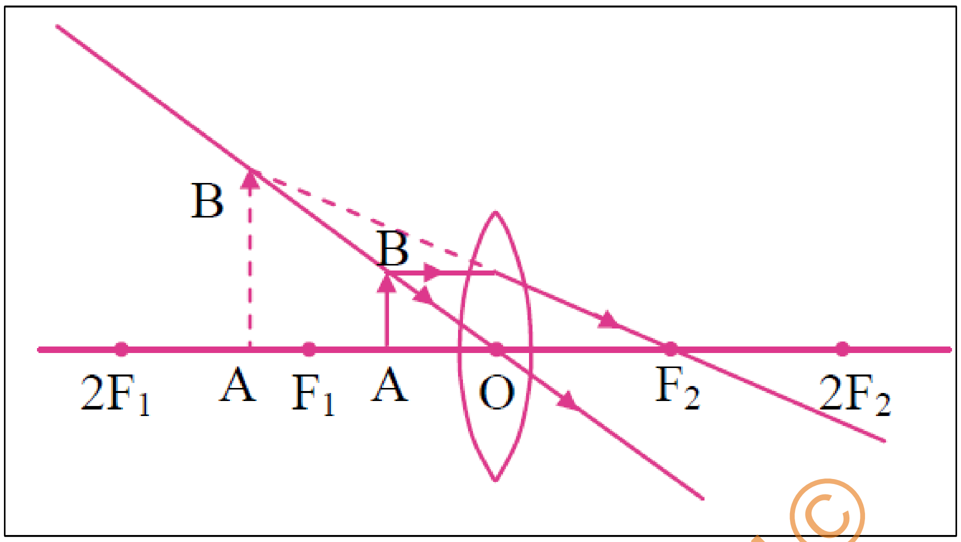
iii. The conical cells respond to the colour and give information about the colour of the object to the brain.

iv. Brain processes all the information received, and we see the actual image of the object.

v. The conical cells can respond differently to red, green and blue colours. When red colour falls on the eyes, the cells responding to red light get excited more than those responding to other colours, and we get the sensation of red colour.

Previous Year Questions

1. Observe the following figure and complete the table:



I. Position of the object: Between F_1 and O

II. Position of the image: On the same side of the lens as the object

III. Size of the image: Very large

IV. Nature of the image: Virtual and erect

2. Write down any two rules used for drawing ray diagrams for the formation of image by convex lens.

Two rules for drawing ray diagrams for the formation of image by convex lens are as follows:

i. When the incident ray is parallel to the principal axis, the refracted ray passes through the principal focus.

ii. When the incident ray passes through the optical centre of the lens, it passes without changing its direction.

3. **Simple microscope is used for watch repairs. Give Reason. – Exercise Question 4 - a**

4. **Write a note on : ‘Persistence of vision’.**

i. We see an object because the eye lens creates its image on the retina.

ii. The image is on the retina as long as the object is in front of us. The image disappears as soon as the object is taken away.

iii. However, this is not instantaneous, and the image remains imprinted on our retina for $1/16^{th}$ of a second after the object is removed.

iv. The sensation on retina persists for a while. This is called persistence of vision.

5. **Surabhi from std. X uses spectacle. The power of the lenses in her spectacle is 0.5 D. Answer the following questions from the given information:**

   I. **Identify the type of lenses used in her spectacle.**

   II. **Identify the defect of vision Surabhi is suffering from.**

   III. **Find the focal length of the lenses used in her spectacle.**

   i. The power of the lenses in her spectacle is positive; hence it is a convex type of lens.

   ii. Since the spectacle has convex lens, Surabhi is suffering from hypermetropia.

   iii. .

   f(m)

6. **Distinguish between hypermetropia and myopia. (3 points)**

7. **Exercise Question 6 - a**

7. **Complete the following table for convex lens:**

At infinity

Same size

Virtual and erect

9. **Draw a scientifically correct labelled diagram of a human eye and answer the questions based on it:**

   a. **Name the type of lens in the human eye.**

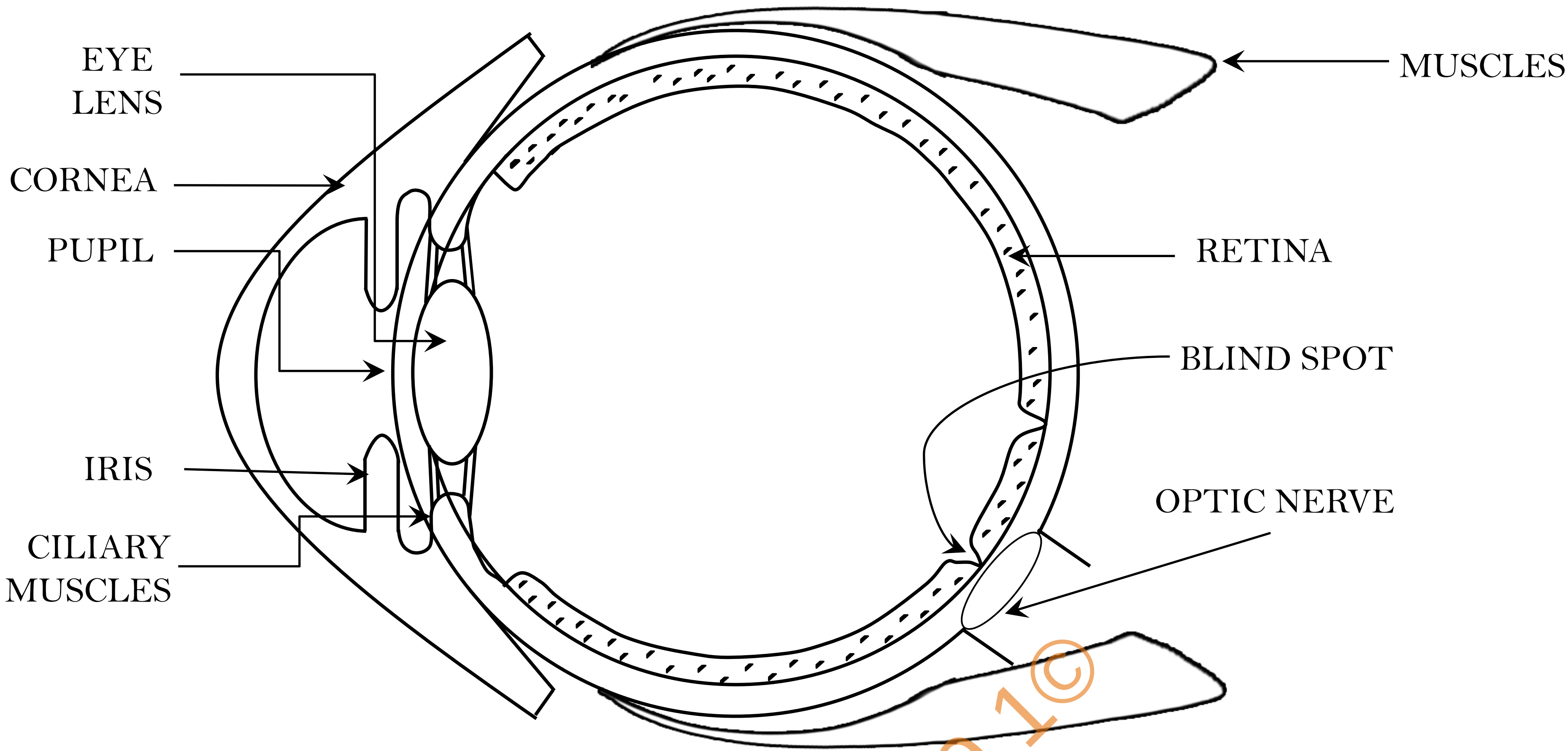
   b. **Name the screen at which the maximum amount of incident light is refracted?**

   c. **State the nature of the image formed of the object on the screen inside the eye.**

a. A double convex transparent crystalline lens.

b. Maximum amount of incident light is refracted at the cornea.

c. Real and inverted


EYE
LENS
CORNEA
PUPIL
IRIS
CILIARY
MUSCLES
MUSCLES
RETINA
BLIND SPOT
OPTIC NERVE